

Anything to submit here?

vibrational theory

Bittner, E. R., et al. (2013). Quantum Biology of the Nose: A Combined Molecular Dynamics and Quantum Chemical Study. *The Journal of Chemical Physics*.

Brookes, J. C., et al. (2007). Could Scent be a Vibration? Quantum Mechanical Mechanism for Olfaction. *Physical Review Letters*.

1. Turin, L. (1996). A Spectroscopic Mechanism for Primary Olfactory Reception. *Chemical Senses*.
 - This paper talks about how olfactory receptors function through inelastic electron tunneling (IET) to detect vibrational frequencies of odorant molecules instead of just their molecular shape
 - This is important because it could mean that quantum mechanical biological mechanism could explain why chemically similar molecules with different vibrations smell different (i.e. isotopes)
 - This is what challenges the current leading lock-and-key theory
 - For my project, this creates the theoretical justification for using vibrational spectra (IR/Raman) as features in a ML pipeline
 - We will need to rigorously find empirical evidence of the efficacy of the predictive modeling because it is a controversial theory
2. Franco, M. I., et al. (2011). Molecular vibration-sensing component in *Drosophila melanogaster* olfaction. *PNAS*.
 - They found that fruit flies can discriminate between deuterated and non-deuterated (a stable isotope of hydrogen) versions of the same molecule — this supports the idea that they can sense molecular vibrations
 - The study is interesting because it changes the debate from theoretical physics to behavioral biology which shows a functional effect of isotopes in animal models
 - This shows “proof of concept” that people cite against the shape-based theory
 - This justifies my inclusions of isotopic or vibrational data in the pipeline because it suggests that biological systems are sensitive to specific shifts
 - Important behavioral evidence keeps vibrational theory viable even though there is a lot of skepticism from molecular biologists

Shape-based theory

Buck, L., & Axel, R. (1991). A Novel Multigene Family May Encode Odorant Receptors: A Molecular Basis for Odor Recognition. *Cell*.

Bushdid, C., et al. (2014). Humans Can Discriminate More than 1 Trillion Olfactory Stimuli. *Science*.

Bozza, T., et al. (2002). Odorant Receptor Expression Defines Functional Units in the Mouse Olfactory System. *The Journal of Neuroscience*.

Dravnieks, A. (1985). *Atlas of Odor Character Profiles*. ASTM Data Series.

1. Block, E., et al. (2015). Lack of evidence for Luca Turin's vibrational theory of olfaction. *PNAS*.
 - By using human odorant receptors expressed in in-vitro systems, they found no evidence that receptors differentiate between isotopes. this means that shape dictates receptor activation
 - This paper is important because it directly tries to frame Turin's model as fundamentally wrong
 - This would be the direct rebuttal to Franco et al. (2011) — which argues that behavioral differences in flies do not exactly imply vibrational mechanisms at a human receptor level
 - This paper is the other side of the argument, supporting lock-and-key mechanism, that I am trying to beat. It sets the standard for why lock-and-key model is currently the leading theory
 - The strongest empirical “no” to vibrational theory — shows the need for a dataset as large as Keller (2017) to find any subtle vibrational signals

machine learning and predictive modeling

Sanchez-Lengeling, B., et al. (2019). Machine Learning for Scent: Learning Generalizable Perceptual Representations of Small Molecules. *arXiv/NeurIPS*.

Gerkin, R. C., & Castro, J. B. (2015). The Number of Olfactory Stimuli that Humans can Discriminate is Still Unknown. *eLife*.

Nozaki, Y., & Hasegawa, S. (2018). Predictive Modeling of Olfactory Percepts with Molecular Vibration Spectra. *Proceedings of the IEEE*.

!! important paper !! (specifically used vibrational spectra to predict DREAM dataset)

Snitz, K., et al. (2013). Predicting Odor Percepts from Molecular Structure. *PLoS Computational Biology*.

1. Sanchez-Lengeling, B., et al. (2019). Machine Learning for Scent: Learning Generalizable Perceptual Representations of Small Molecules. *arXiv/Google Research*.
 - This paper uses graph neural networks (GNN) to predict odor properties from molecular structures with tones of accuracy from the DREAM dataset
 - it is significant because it moves the focus from manually engineering features (shape or vibrations) to learned representations of the entire molecular graph
 - It builds directly from the Keller et al. (2017) DREAM challenge — winning the benchmark that my project is aiming to challenge

- This study acts like a “gold standard” for the shape pipeline. this represents that most intricate way to extract features from 2D/3D geometry
- Important reminder that for relevancy of vibrations, they have to have information that a deep neural network can't already extract from the molecular graph

technological and molecular sensing

Mainland, J. D., et al. (2014). Human Olfactory Receptor Responses to 511 Odorants. Scientific Data.

Saito, H., et al. (2009). Odor Coding by a Mammalian Receptor Repertoire. Science Signaling.

1. Nozaki, Y., & Hasegawa, Y. (2018). Predictive Modeling of Odor Character of Molecules from Their Molecular Vibrational Spectra. Analytical Chemistry.
 - They tested if machine learning models trained on IR spectra can successfully predict certain odor qualities like “fruity” or “minty”
 - This is delicate because it relies on the correlation between functional groups (specific vibrations) and odors (doesn't exactly prove the vibrational mechanism)
 - Bridges the gap between Turing's theory and the DREAM challenge by applying current ML to the specific data type (vibrations) I am looking into.
 - This gives me the previous research which shows that vibrational spectra has enough information to predict odor even if the biological mechanism is debated
 - an important technical precedent that proves the “shake” data is informative regardless of whether the nose is a spectroscope or not